GATE

ALL BRANCHES



General Aptitude

ANALYTICAL APTITUDE

Lecture No.- 01



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Recap of Previous Lecture



Topic

Interest

Topics to be Covered

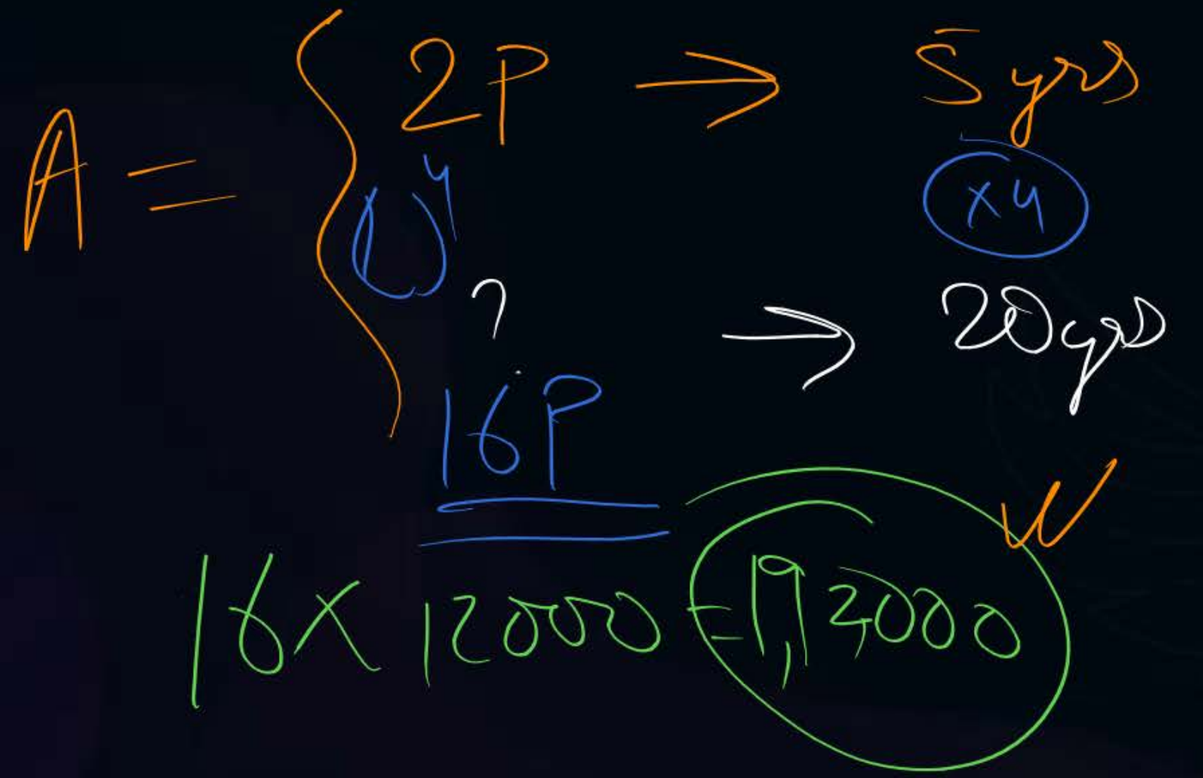


Topic-1 Blood Relations

[MCQ]

#Q. A sum of ₹12000 deposited at compound interest becomes double after 5 years. After 20 years, it will become?

Assignment



- A sum of ₹12000 deposited at compound interest becomes double after 5 years. After 20 years, it will become?

- Sol: The rate of interest at which ₹12000 doubles after 5 years is given by
- $R = 100(x^{\frac{1}{5}} - 1)$
- $= 100(2^{\frac{1}{5}} - 1)$
- $=100 \times (1.1486 - 1)$
- $= 100 \times 0.1486 = 14.86\%$

Approx

Therefore, after 20 years it becomes,

$A = ₹12000(1 + 14.86/100)^{20} = ₹12000\ (1.1486)^{20}$

$= ₹12000 \times 15.97 \quad = ₹\ 191671.474$

[MCQ]

#Q. What is the compound interest on ₹2500 for two years at the rate of interest 4% per annum?

$$C.I = P\left(1+\frac{R}{100}\right)^{T}$$

Assignment

A ₹ 180

B ₹ 210

C ₹ 204

D ₹ 220

$$2500 \times 1.04 \times 1.04 = 2704$$

$$I = 204$$

[MCQ]

$R = x\%$ $\quad NR = (x+4)\%$

#Q. A sum of money was invested at simple interest at a certain rate for 3 years. Had it been invested at a 4% higher rate, it would have fetched ₹480 more. Find the principal.

Assignment

A ₹ 3000

B ₹ 4000

C ₹ 5000

D ₹ 6000

$4\% \; ? \rightarrow 160$

$100\% \; ?$

$\frac{160}{4} \times 100 = 4000$

$\frac{480}{3} = 160$

[MCQ]

$1.06 \times 1.06 \times 1.06 \ldots\ldots$ 2

#Q. In what time a sum of money will double itself at a rate of S.I. of 6% per annum?

Assignment

$6 \times ? = 100$

$? = \frac{100}{6} = \frac{50}{3} = 16\frac{2}{3}$ yrs

A. 6 ⅓ years

B. 16 ⅔ years

C. 33 ⅓ years

D. 21 years

[MCQ]

#Q. In how many years will a sum of ₹800 at 10% per annum compounded semi-annually become ₹926.10?

Half Yearly

Assignment

A 1 ⅓ years

B 1 ½ years

C 2 ⅓ years

D 2 ½ years

$P = 800$ | $R = 10\%$ | years | $T = ?$

$R = 5\%$ | Half Yearly

$A = P\left(1+\frac{R}{100}\right)^n$

$926.10 = 800\left(1+\frac{5}{100}\right)^n$

$\frac{9261}{8000} = \left(\frac{105}{100}\right)^n = \left(\frac{21}{20}\right)^n$

$\Rightarrow \left(\frac{21}{20}\right)^3 = \left(\frac{21}{20}\right)^n$

$\therefore n = 3$

$T = 1\frac{1}{2}$ year

[MCQ]

S.I $\Rightarrow 0.16 \times 2500 = 400$

#Q. The difference between simple interest and compound interest on a sum for 2 years at 8% when the interest is compounded annually is ₹16. If the interest were compounded half yearly the difference in two interests would be nearly

A ₹ 21.25

B ₹ 24.64

C ₹ 27.85

D ₹ 29.94

$\text{Diff} = P\left(\frac{r}{100}\right)^2$

$16 = P\left(\frac{8}{100}\right)^2$

$\Rightarrow 16 = P \times \frac{4}{625}$

$\Rightarrow \frac{16 \times 625}{4} = P = 2500$

Diff 24.64

CI $\Rightarrow$ 424.64

Assignment

$A = P\left(1+\frac{r}{100}\right)^4$

$= 2500\left(1+\frac{4}{100}\right)^4$

$= 2500 \times 1.04 \times 1.04 \times 1.04 \times 1.04$

$= 2924.64$

[MCQ]

#Q. Raju invested Rs.5488 for 3 years at C.I. and received an amount of Rs.6750 on maturity. What is rate percent?

Assignment

A. $7\frac{1}{7}\%$

B. $9\frac{2}{3}\%$

C. $3\frac{1}{3}\%$

D. $6\frac{1}{3}\%$

Blood Relations

Great G.F / Great G.M

Forefather

Mother / Father

Brother / sister

Uncle / Aunt

G.F / G.M

Grand son / Grand daughter

Paternal
Maternal

Cousins

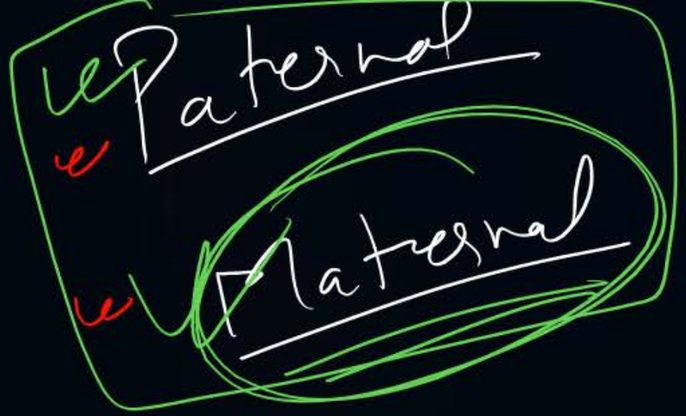
Nephew Niece

Son daughter

Husband wife



Blood Relations



(Parent)

(Child)

Mother : Father
Brother : Sister
Uncle : Aunt
Grand Father : Grand Mother
Grand son : Grand daughter

(Sibling)

(Grand parent)

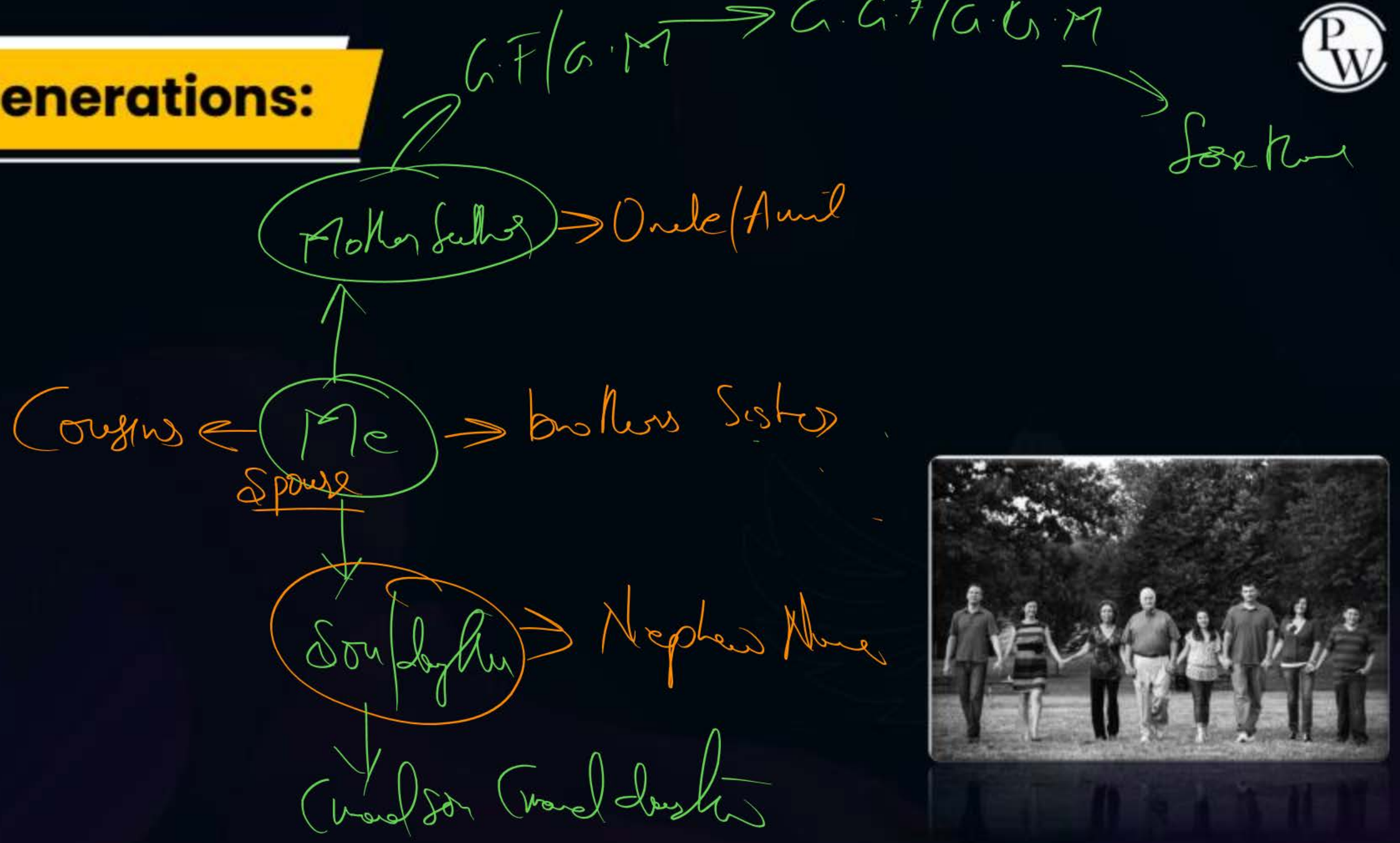
Son : Daughter
Cousins
Nephew : Niece
Husband : Wife

(Grand Child)

(Spouse)

Great Grand Father : Great Grand Mother
Forefather / Ancestors

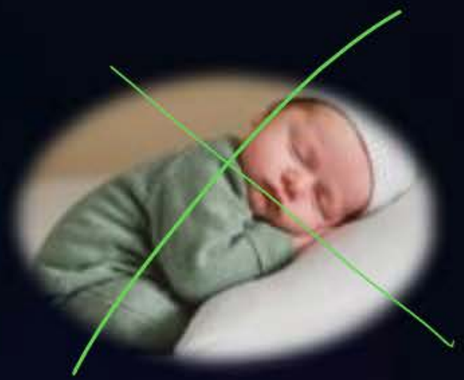
Generations:



In laws:

Son in law
Daughter in law

Father in law
Mother in law
Brother in law
Sis in law



before



After

In laws:

Spouse Mother = Mother in law
Spouse Father = Father in law
Spouse Brother = Brother in Law
Spouse Sister = Sister in law

Son in Law
Daughter in Law

~~Spouse son~~ = Son
~~Spouse daughter~~ = daughter

[MCQ]

#Q. Pointing to a boy in the photograph Manasa said, "His sister is the only daughter of my father". How is the boy related to Manasa's father?

A Father

B Brother

C Son

D Cousin

[MCQ]

#Q. Pointing to Neha, Mahesh said, "Her mother's only daughter is my daughter". How is Mahesh related to Neha?

A Brother

B Son

C Uncle

D Father

[MCQ]

#Q. A is brother of B. B is wife of C. C is son of D. D is wife of F.

Then what is B to F?

Assignment

A Daughter-in-law

B Cousin

C Son-in-law

D None of these

[MCQ]

#Q. Pointing towards a person in a photograph, Raman said, "She is the only daughter of the mother of my brother's sister". How is that person related to Raman?

Assignment

A. Father

B. Mother

C. Uncle

D. Sister

[MCQ]

#Q. If Neena says, "Anita's father Raman is the only son of my father-in-law Mahipal", then how is Bindu, who is the sister of Anita, related to Mahipal?

Assignment

A Daughter

B Wife

C Daughter-in-law

D None of these

[MCQ]

#Q. Pointing to a photograph, a woman says, "This man's son's sister is my mother-in-law." How is the woman's husband related to the man in the photograph?

Assignment

A. Son

B. Grand son

C. Son-in-law

D. Nephew

[MCQ]

#Q. When Amit saw Manoj, he recalled and said "He is the son of the father of the mother of my daughter". What is Manoj to Amit?

Assignment

A Cousin

B Brother-in-law

C Uncle

D Brother

[MCQ]

#Q. Direction:

X+Y means X is the father of Y,
X-Y means X is the wife of Y,
X × Y means X is the brother of Y,
X ÷ Y means X is the daughter of Y.

Assignment

A x B - C means _____

A. A is husband of C

B. A is brother-in-law of C

C. A is son of C

D. Can't be determined

[MCQ]

#Q. Direction:

X+Y means X is the father of Y,
X-Y means X is the wife of Y,
X × Y means X is the brother of Y,
X ÷ Y means X is the daughter of Y.

Assignment

In A x B ÷ C, which of the following statement is true?

A. A is father of C

B. A is brother of C

C. A is uncle of C

D. A is son of C

[MCQ]

#Q. Rohit said to Mohit, "The boy playing basket ball is younger brother of the two brothers of the daughter of my father's wife". How is the boy playing basket ball related to Rohit?

Assignment

A Brother-in-law

B Nephew

C Cousin

D Brother



2 mins Summary

Topic Blood Relation

THANK - YOU